

Co-Orb: Fostering Remote Workplace Gratitude with IoT Technology

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Figure 1: Co-Orb: A smart IoT device that brings thank-you messages from screens to life.

ABSTRACT

While gratitude in the workplace is essential for group well-being and teamwork efficiency, showing gratitude to coworkers becomes difficult under remote work settings, which is prevalent with the ongoing pandemic. Our research revealed that in remote workspaces, people often feel less emotionally touched by the appreciation messages sent by coworkers via text-based communication due to the lack of in-person interaction. In addition, it has become more challenging for coworkers to express their appreciation in a sincere and

timely manner. To solve this problem, we designed Co-Orb, an IoT desktop orb that is connected to people's work communication app and lights up with a customized emoji when a thank-you message is received from co-workers. By creating a multi-sensory experience that brings thank-you messages beyond screens, we aim to make it more enjoyable for users to express and receive appreciation in a remote workplace, thereby promoting better work relationships.

CCS CONCEPTS

• Human-centered computing → Collaborative and social computing systems and tools.

KEYWORDS

Remote work; gratitude; IoT; interaction design; co-worker relationships; workplace; collaboration

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1 INTRODUCTION

Feeling valued is one of the most fundamental needs of humans, particularly in the workplace where mutual recognition plays a key role in healthy work relationships. Research shows that gratitude in the workplace, which mostly stems from appreciation between co-workers, not only increases people's job satisfaction, but also fosters team learning and promotes group well-being [1]. However, as COVID-19 pushes us to work remotely, it has become particularly challenging to express and perceive gratitude during work due to the lack of face-to-face interactions.

When people work in the office, they can easily express their gratitude to their co-workers by handing them a cup of coffee or inviting them for lunch. These are personalized, timely and specific enough to deliver their emotions [12]. Unfortunately, when it comes to remote work, those simple but effective interactions are no longer an option.

Our research dug deep into the non-verbal interactions between co-workers in remote work settings. In order to eliminate the nuanced problems among remote co-workers, we designed "Co-Orb", a smart IoT device that brings thank-you messages from the virtual world to reality. We envision that Co-Orb can promote positive connections between co-workers and improve group well-being in remote workplaces by encouraging the expression of gratitude and amplifying the perception of gratitude.

2 INITIAL RESEARCH

2.1 Literature Review

Many past researches suggest that gratitude in the workplace is positively correlated with work satisfaction and employees' well-being. People's level of happiness could be increased by performing acts of kindness and gratitude [6]. Gratitude interventions can be utilized as a way for team leaders to boost such job satisfaction by "regularly prompting grateful emotions" among coworkers [14]. In addition, most employees prefer verbal one-on-one gratitude to any other forms, and timeliness and specification are believed to have tight associations with an enhanced sense of gratitude [3].

In a remote work setting, however, these key qualities normally associated with sincere gratitude expression have become harder to achieve, leading people to feel unacknowledged and less connected with other co-workers. Furthermore, the communication of emotions, including gratitude, largely relies on visual and auditory cues that are better presented in face-to-face interactions [8]. However, those cues are generally missing in the main channels of communications in a remote work setting, such as text messages and emails, which poses additional challenges to the successful delivery of grateful emotions between co-workers. Thus, it is important for us to consider alternative methods beyond text-based messages to mitigate the negative impact of remote communication on workplace gratitude.

Many previous researches explored the possibility for distant communication of emotions via tangible objects [2]. For example, LumiTouch, an interactive picture frame presented by MIT Media

Lab, showed that ambient display could serve as a good tool to communicate emotional information without interfering with people's focal attention [5]. These researches provide a good foundation for our exploration.

2.2 Desktop Research

We conducted desktop research on existing products and features that target the realm of gratitude expression in workplace and team-work settings. HeyTaco, for instance, is a Slack-integrated app that adopts gamification to promote recognition and grateful expression amongst team members [7]. The company Yelp also has an internal system called Yelp Love, in which employees can send hearts to show appreciation to colleagues [13]. These existing products serve as great inspirations for our research and design. Meanwhile, we also recognized their limitations in remote work settings where all the interactions are done over the computers, which makes it challenging for individuals to fully receive the grateful emotions behind the screens. Therefore, we devoted our subsequent research efforts to uncovering ways that can help employees who work remotely to better perceive gratitude from colleagues when in-person interactions are missing.

2.3 Semi-Structured Interview

To understand how people perceive and show gratitude to coworkers, as well as associated problems they could encounter in remote work settings, we conducted 10 semi-structured interviews using directed storytelling methods. Participants were recruited from social media platforms and were asked to complete the consent form before participating. We created a screener to make sure the representative users coming from various backgrounds, aged between 18 to 60, who had at least two-month long remote-work experiences during the pandemic.

Here're some interview questions we asked during the interview:

- (1) Can you recall the most recent time you received appreciation from your co-worker during remote work? How do you feel about it?
- (2) Can you recall the most recent time you gave appreciation to your co-worker during remote work? How do you feel about it?
- (3) Are there any differences in receiving/giving appreciation between remote-work and in-person work?
- (4) In general, when would you like to give/receive appreciation?
- (5) What way of showing appreciation do you prefer and why?

After a note interpretation session and three rounds of affinity diagramming, we synthesized the insights from the interviews. We found that without in-person interactions:

- (1) People feel less emotionally touched by the appreciation they receive from co-workers, which in most cases are text messages from the work communication platforms.
- (2) People find it difficult to show their appreciation to others in a sincere and timely manner.
- (3) People feel that they did not receive as much appreciation as they expected.

These are two participant quotations that we captured in type-written notes during the interview sessions:

“For me, seeing a thank you message on Slack feels less emotionally touching than hearing someone say it in person. It almost feels less sincere if I don’t get to see that person’s facial expression...”

“You know if we work together and the person just sits next to me, I can just pat on his shoulder and hang out for lunch to thank him. But now, I type a “thank you” and tell myself to thank him in other ways afterward... I might end up forgetting this cause I don’t see him in person...”

Based on these findings, we decided on our design focus: developing a solution that makes it convenient and enjoyable to express and perceive gratitude amongst co-workers.

3 DESIGN PROCESS

3.1 Ideation

To brainstorm initial ideas, our team held a Crazy 8’s¹ session and generated 32 ideas. After a round of discussion, our team ranked the 4 most preferred ideas that could best tackle our target problem while highly practical in real life. To gain initial feedback from potential users, we drew 4 storyboards and performed speed dating sessions with 12 participants who are currently working in remote settings (Fig. 2). We organized the speed-dating results (Fig. 3) using sticky notes and synthesized them using affinity-diagrams (Fig. 4).



Figure 2: Speed Dating with Storyboards

Feedback from users showed that 1) they favor solutions that do not take too much effort. 2) they prefer solutions that are not solely screen-based. Therefore, we decided to go with the Co-Orb concept: an IoT desktop orb that connects to people’s work communication app and lights up every time an appreciation message is received during work.

¹Crazy 8’s: each team member draws 8 ideas within 20 minutes for quick ideation.

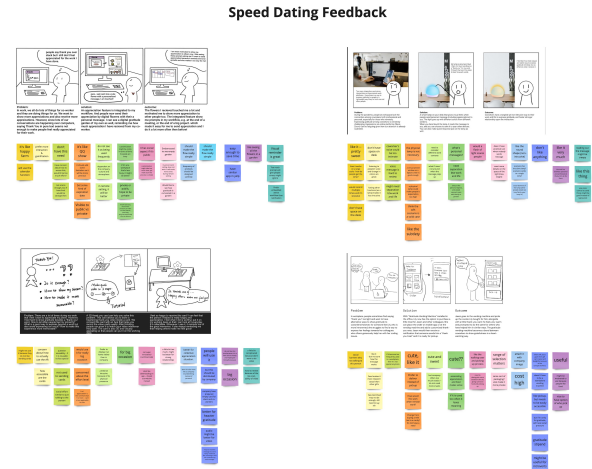


Figure 3: Speed Dating Results

Speed Dating Insights

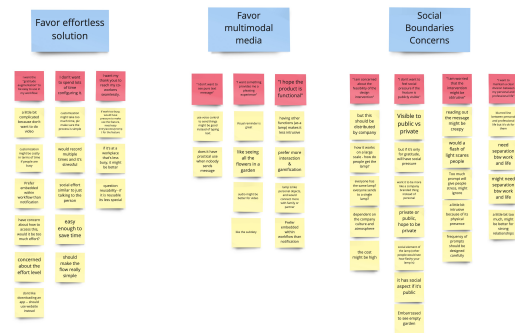


Figure 4: Speed Dating Results

3.2 Initial Design

The initial design was divided into 2 main parts: a plugin application to encourage the expression of appreciation, and an IoT orb on people’s desktops to receive appreciation.

3.2.1 To express appreciation. Based on what we learned from previous speed dating workshops, sending appreciation to co-workers should be effortless and well-integrated into people’s daily workflow. In our design, we developed a plugin app that could be integrated with people’s work communication platforms. We selected Slack, a commonly used work communication platform to build our prototype upon. When a user sends a message that contains “thank”, “appreciate” or other keywords that are related to gratitude, a prompt would show up in Slack and ask if the user would like to light up the receiver’s Co-Orb.

3.2.2 To receive appreciation. When a user receives an appreciation message from a co-worker, the orb, placed on their work table, will gradually glow up for several seconds. Users can then tap their hands on the orb to play a synthesized voice reading out the message. The orb displays a pastel yellow light when a message is received to convey a sense of warmth and enhance the delightfulness of the sensory experience, thereby making each message more memorable and emotionally touching.

3.3 User Testing and Iterations

To test our design concept, we ran Wizard of Oz testing² sessions with 8 target users using think-aloud protocol (Fig. 5). Each user conducted two tasks during the test guided by one facilitator. During the first part, the user was asked to sit in front of the working computer and went through the experience of receiving an appreciation message via a physical orb next to the computer. Next, the user was asked to send out an appreciation message to one co-worker in two different ways using the digital Figma prototype. Besides, the user was also encouraged to play around and navigate through the whole app. Upon completing, the facilitator followed up with some interview questions depending on the user's behavior during the test.

We received a lot of positive feedback from participants; many said that they felt emotionally touched and would be more motivated to express gratitude to coworkers if they were to use the orb in their everyday work. However, the testing process also revealed some blind spots in our design and indicated opportunities for improvement.



Figure 5: User testing using Wizard of Oz method

3.3.1 Iteration 1: Make Appreciation Messages Personalized and Collectable. A key piece of feedback we received from participants was that adding a bit more personalization might make the lighting-up experience more special and memorable. Based on this feedback, we added a function that allows users to pick an emoji when lighting up the orb. On the other side, the receiver will see the emoji

²Wizard of Oz is a method where participants interact with a system that they believe to be autonomous, but in reality, is controlled by an unseen human operator in the next room.

displayed on the surface of the orb when it lights up. A fair number of users also mentioned that they wanted to collect the appreciation messages they received. To improve on this, we designed a web page where users could review all the appreciation messages they received.

3.3.2 Iteration 2: Reinforce the Principles of Calm Technology. Some users pointed out that they hope this orb does not consume too much of their attention away from work. This feedback confirmed our belief that this design should follow the principles of Calm Technology, which means that it should create and inform calmness and only draw users' attention when it is needed [4]. To reinforce these principles in our design, we made three improvements:

- (1) Show appreciation sender's name on the orb when it lights up. This change allows the information to be more glanceable and more accommodating for different scenarios, for example, when the recipient is busy with work.
- (2) Allow users to customize the time they would like to check the gratitude message via the orb. For example, users could set to check the orb only when they finish the work or every several hours to avoid distractions from work.
- (3) Have the orb light up in a more gentle way to avoid obtrusiveness and enhance calmness.

4 FINAL DESIGN

4.1 Send & Receive Gratitude

To foster gratitude among co-workers in remote work settings, we designed Co-Orb, a smart IoT device at every co-worker's home office to exchange gratitude in a more tangible format (Fig. 6). An appreciation message from a team communication platform will light up the user's orb with a customized emoji and the sender's name, inviting the user to touch. After touching the orb, the content of the appreciation message will be read out, mimicking the experience of hearing it in person.



Figure 6: Co-Orb: A Smart IoT Device

Sending such appreciation messages to a coworker's Co-Orb is done through a plugin app embedded in work communication

platforms (Fig. 7). Once installed, the plugin app will automatically detect keywords associated with gratitude in the user’s messaging threads and prompt the user to light the recipient’s Co-Orb when a keyword is detected. If the user wants to opt out of auto-detection due to privacy concerns, they can utilize our dedicated chatbot, which will recommend coworkers to express more appreciations based on the frequency of their interactions (Fig. 8).

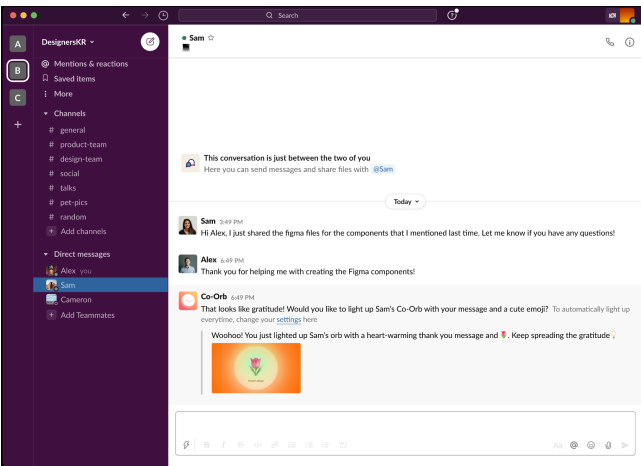


Figure 7: Co-Orb: Plugin App on Slack

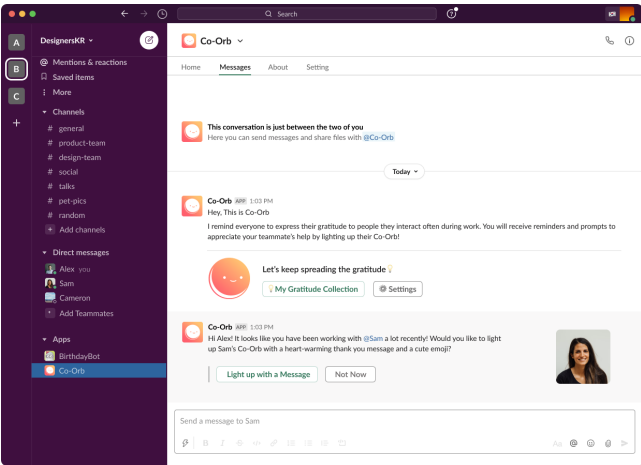


Figure 8: Co-Orb: Recommending Coworkers to Thank

4.2 Collect Gratitude

To review all the previous appreciation messages people have received, they can visit their personal Co-Orb’s page which showcases the gratitude collection gathered from Co-Orb (Fig. 9). The collection page visualizes all appreciation messages using Emoji as an entry rather than names or timestamps, which makes the message-retrieving experience similar to the gratitude moment on the orb.

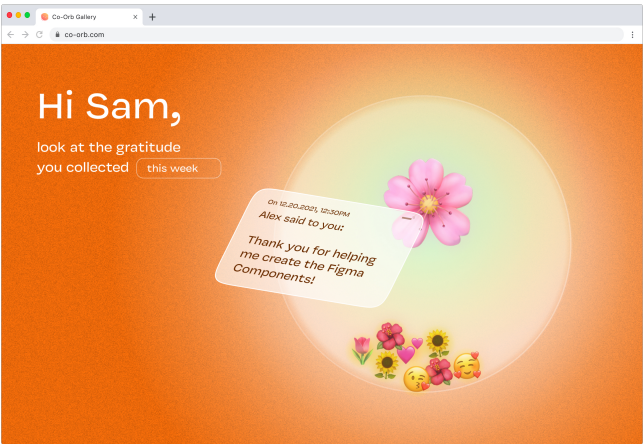


Figure 9: Co-Orb: Gratitude Collection Page

4.3 Privacy & Preferences

In a work-from-home setup, Co-Orb sits in between a communication device and a smart home device. This is why giving the users full access to customization is important to help them to find a preferred balance. Co-Orb provides a comprehensive setting page (Fig. 10) with a variety of options, ranging from adjusting the color of the light to opting out from features deemed a bit intrusive for their taste.

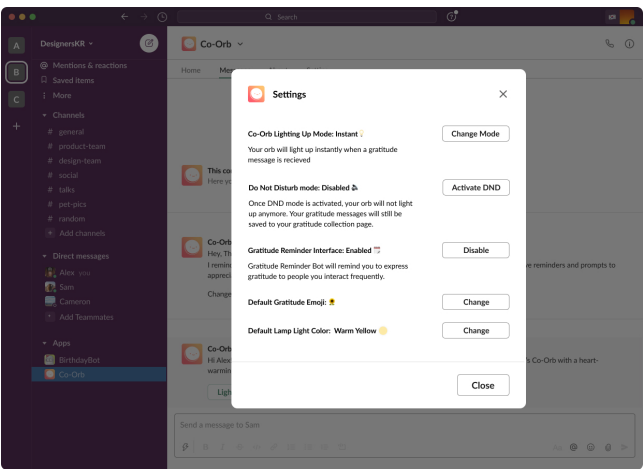


Figure 10: Co-Orb: Preference Setting Page

Privacy concerns might arise from chat monitoring and message readout. Firstly, we conceive of the gratitude recommendation (Fig. 8) feature as an optional reminder for users who are not used to expressing gratitude. If chat monitoring is turned off, our dedicated chatbot for appreciation messages is an easy workaround to gratitude recommendation. Message readout only happens by default upon touching. Additionally, we intend to add a physical mute button on the device to give extra control over cases, such as overhearing. Since Co-Orb makes texts and emojis more tangible as a tool for communication, we encourage users to get creative with

their expression. However, in cases of abuse, such as messaging with offensive content, users can choose to block specific senders. This will save the troubles of disturbing messages being read and overheard.

4.4 Feasibility & Cost

During the Wizard of Oz testing, we utilized projection mapping onto a normal lamp to showcase the experience to our users. For real production, the product could be fully implemented with a spherical lamp powered by a led display and a Raspberry Pi. The orb display will be mounted right on the inside of the spherical frosted acrylic enclosure to ensure both even and clear pass-through lighting. The potential technologies for implementation are spherical LED panels [10], spherically arranged LED strips [11], or stacked circle-shaped PCB with LED pixels [9].

To promote an organization-wide usage of Co-Orb, we imagine that it will be purchased and distributed by companies to their employees as part of a home-office kit. Based on the production plan above, we estimate that the cost of Co-Orb can be maintained at a reasonable level and can be outweighed by the numerous benefits that Co-Orb can bring to an organization. While we expect the benefits of Co-Orb to reach their maximum in the context of an organizational usage, we also realize that a large-scale adoption can be challenging due to various reasons, such as a divergence in preferences from individual employees. To address the possible cases in which employees are reluctant to use Co-Orb, we propose a digital analogous form (Fig. 7) to the physical orb as an alternative channel to get the gesture of lighting up across without disrespecting personal preferences. In that case, when one employee sends an appreciation message to a colleague who has chosen not to use the physical orb, the Co-Orb bot will auto-generate and send a message to the recipient that has an animation embedded to mimic the physical experience.

Another consideration in our intervention is the potential impact on the balance between personal and professional life for employees. Because we expect the usage of Co-Orb to be in a home office setting, one factor to consider is how employees will perceive and interact with Co-Orb outside of their regular work time. To avoid misconceptions from employees regarding the company's intention of distributing Co-Orbs, we propose an additional feature that aims to avoid the unintentional encouragement of overtime work and invasion of personal space. In the settings page, users can specify their work hours and off time, and turn on the "do not disturb" option to prevent Co-Orbs from lighting up when they are off work. In that case, all messages sent outside of the designated work hours will be saved and become available for retrieval on the next working day.

5 CONCLUSION AND FUTURE WORK

Powered by IoT technologies, Co-Orb is designed to foster gratitude among co-workers in remote work settings. During user testing, using Co-Orb to send, perceive, and collect gratitude has shown to be an enjoyable and touching experience for our target users. Participants mentioned that this product brought them unexpected heartwarming feelings and would motivate them to express more gratitude at work. Due to the limitation of time, we were only able

to conduct user testing with 8 participants under simulated remote work scenarios. Because of this constraint, we did not get to see the extent of engagement over time, which is one major limitation of our study. For future work, we plan to 1) evaluate our product via longitudinal tests to examine the long-term impact of Co-Orb on workplace gratitude 2) pitch Co-Orb to local companies and seek collaboration opportunities to build and test the product with end-users in a real workplace environment 3) extend the compatibility of the product by supporting the integration with more work communication platforms.

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